Serena A. Cronin

PhD Candidate, Astronomy | University of Maryland, College Park | cronin@umd.edu | serenacronin.github.io
ORCID: 0000-0002-9511-1330 | UAT: Galaxy evolution (594), Galaxy winds (626), Starburst galaxies (1570)

EDUCATION

University of Maryland — College Park, MD
Ph.D. Astronomy — Expected May 2027
M.S. Astronomy — May 2023

The Ohio State University — Columbus, OH
B.S. Astronomy and Astrophysics — Aug. 2016 – Dec. 2020
Cum Laude with Research Distinction

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Maryland — Aug. 2021 –
Dissertation: A Tale of Two Starbursts: The Multiphase Galactic Winds of M82 and NGC 253
Advisor: Prof. Alberto Bolatto

National Radio Astronomy Observatory Intern — May 2020 – Aug. 2021
*Project: Molecular Gas Toward the Supermassive Black Hole Sgr A**
Advisors: Drs. Juergen Ott, David Meier, and Brian Svoboda

Undergraduate Research Assistant, Ohio State University — Aug. 2018 – Aug. 2021
Thesis: Local Environments of Low-Redshift Supernovae
Advisors: Prof. Adam Leroy, Dr. Dyas Utomo

SELECTED OBSERVING

*I have led the design, proposal, and execution of **four PI observing programs** using the JWST, VLA, and ALMA, totaling 130 hours of awarded time, in addition to a substantial primary-observer role for the GBT.*

JWST

PI – Cycle 5, **9 hr**, NIRSpec MSA — ID: 10482
Crushing It: Tracking the Survival of Cool Clouds on Cloud-Crushing and Dynamical Timescales in the M82 Outflow

VLA

PI – Semester 26A, Rank A, **87 hr** — ID: 26A-453
Saying 'HI' to the Deepest View of M82's HI Halo

ALMA

PI – Cycle 12, Rank C, **27 hr** — ID: 2025.1.00928.S
Five Golden Rings: Super Star Cluster Formation in the Nuclear Rings of Five Nearby Galaxies

PI – Cycle 11, Rank A, **7 hr** — ID: 2024.1.00917.S
DUVET-Keck: A Sub-kpc View of Star Formation Regulation Across Local Starbursting Disks

GBT

Co-I – 2022–2025, PI: Alberto Bolatto — ID: 21B-024
GBT EDGE: A Representative Survey of the z=0 Universe with Full IFU Spectroscopy

- Primary Observer: 48.5 hours
- Designed observations: 123.5 hours

GRANTS & AWARDS

- JWST-GO-10482, JWST/STScI/NASA – $123,512 2026
- Andrew S. Wilson Prize for Excellence in Research, Dept. of Astronomy, UMD – $500 2026
- Department Service Prize, Dept. of Astronomy, UMD – $500 2023
- Philip E. Angerhofer Outstanding Teaching Assistant Award, UMD – $250 2022
- AAS 238 Chambliss Astronomy Achievement Student Award – Honorable Mention 2021
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- Undergraduate Research Grant, Ohio State – $1,500 2019
- Astronomy Undergraduate Scholarship, Ohio State – $1,000 2019

MENTORING, LEADERSHIP, & SERVICE

Research Mentoring

Primary research mentor to six students, leading project design and implementation; outcomes include a co-authored publication, a mentee now pursuing a Ph.D., and letters of recommendation supporting applications.

Joshua Rickford	2026-	University of Maryland
Alanis Alvarado	Summer 2025	GRAD-MAP Summer Scholars
Angel Rodriguez-Barbosa	Summer 2024	University of Maryland
Rhiannon Paterson	Summer 2024	GRAD-MAP Summer Scholars
Tanya Bait	2023−2024, 2026-	UMD Honors Thesis, STEM Magnet Program
Lenin Nolasco*	Summer 2023	GRAD-MAP Summer Scholars

*Co-author, Cronin et al. 2025, ApJ, 987, 92

Committee & Service Roles

- **Referee** – The Astrophysical Journal (ApJ)
- **NASA ROSES Executive Secretary** (April 2025)
- **American Astronomical Society SGMA Committee** (2024–present)
- **Community and Belonging Committee**, UMD (2022–2024)

Founder & Program Leadership

- **Workshops on Applying to Astronomy Graduate School**, Founder & Organizer (2023–2025)
  Built and led a UMD-based workshop series teaching undergraduates the graduate-school application process, including CV and essay workshops.
- **Astronomy and Physics Pride Alliance**, Founder & Organizer (2022–2026)
  Founded and led a community-building group for students in physics and astronomy at UMD.

Additional Outreach & Service

- **Total Solar Eclipse Outreach Event**, Organizer, Cleveland, OH (Apr. 2024) – solar observing and demos
- **Python Bootcamp Lead**, GRAD-MAP Winter Workshop, UMD (Jan. 2024)
- **Graduate Peer Mentoring Program**, UMD (2022–2023)
- **Maryland Day Volunteer**, UMD (Apr. 2022, 2024, 2025) – solar observing, spectroscopy demos

PRESENTATIONS

⋆ *invited talk* ○ *talk* △ *poster*

○ TUNA Lunch Talk	*The Evolution of Entrained Cool Material in Starburst-Driven Galactic Winds Revealed by JWST*	NRAO (VA)	Oct. 2026
○ Exploring the Aromatic Universe in the Era of JWST	*PAH Survival in the M82 Superwind with JWST*	London, Ontario	July 2026
⋆ Low Density Universe Seminar	*Dust Processing in the M82 Wind with JWST*	STScI	April 2026

◦ UKansas Astro. & Space Physics Seminar	*Dust Processing in the M82 Wind with JWST*	Virtual	Mar. 2026
◦ Dusty Universe: 5th Pan-Dust Conference	*Dust Survival in the M82 Wind with JWST*	UArizona	Nov. 2025
⋆ CMNS Board of Visitors Meeting	*Surviving the Storm: Tracing Cold Material in Hot Galactic Winds with JWST*	UMD	Oct. 2025
△ Star Formation, Stellar Feedback, and the Ecology of Galaxies	*Dust Processing in the M82 Superwind with JWST*	Visegrad, Hungary	May 2025
⋆ Galaxy Journal Club	*Ionized Gas Measurements of the NGC 253 Superwind with VLT/MUSE*	STScI	Feb. 2025
⋆ JWST Weekly Briefing	*Dust Processing in the Iconic M82 Wind with JWST*	STScI	Jan. 2025
◦ AAS 245	*Dust Processing in the Iconic M82 Wind with JWST*	Washington, DC	Jan. 2025
◦ UKansas Astro. & Space Physics Seminar	*Ionized Gas Measurements of the NGC 253 Superwind with VLT/MUSE*	Virtual	Nov. 2024
△ Green Bank Observatory	*Observations of the Ionized NGC 253 Wind with VLT/MUSE*	Green Bank, WV	June 2024
△ AAS 238	*Molecular Gas Toward the Supermassive Black Hole Sgr A**	Virtual	June 2021
△ AAS 237	*The Local Environments of Low-Redshift Supernovae*	Virtual	Jan. 2021
◦ NRAO Summer Student Symposium	*What Surrounds the Supermassive Black Hole Sgr A*?*	Virtual	Aug. 2020
△ Denman Undergrad. Research Forum	*The Local Environments of Low-Redshift Supernovae*	OSU	Mar. 2020
△ APS Conf. for Undergrad. Women in Physics	*The Local Environments of Low-Redshift Supernovae*	Carnegie Mellon	Jan. 2020
◦ Summer Undergrad. Research Program Symposium	*The Local Environments of Low-Redshift Supernovae*	OSU	July 2019

TEACHING

- **Teaching Assistant**, ASTR 100: Introduction to Astronomy, Univ. of Maryland — Fall 2021 – Spring 2022
- **Course Development**, ASTRON 1101: From Planets to the Cosmos, Ohio State — Spring 2021
- **Teaching Assistant**, ASTRON 1101: From Planets to the Cosmos, Ohio State — Spring 2017 – Fall 2020

SKILLS & PROFESSIONAL DEVELOPMENT

Programming Languages Python, C++
Technical Skills CASA, Linux, LaTeX, Wolfram Language, HTML
Workshops ALMA Data Reduction Workshop (2024); Green Bank Single Dish Summer School (2024); JSI Workshop on Winds throughout the Universe (2023); Green Bank Observatory Remote Observer Training (2022); NRAO 18th Synthesis Imaging Workshop (2022)

PRESS & MEDIA

- Featured in "GRAD-MAP Students, Mentors Learn From Each Other" — August 2025
- Featured in "Studying Dust in the Wind" — June 2025
- Quoted in NASA press release: "NASA's James Webb Space Telescope Captures Images of Galaxy M82's Glowing Core and Galactic Wind" — April 2024

PUBLICATIONS

In Prep. First Author

1. *JWST Observations of Starbursts: Polycyclic Aromatic Hydrocarbons in the NGC 253 Superwind*
   Cronin, Serena A., et al., to be submitted to ApJ Letters Nov. 2026

Total: 18 papers — 208 citations — h-index = 9 — SciX (ADS) Library
First-authored: 3 papers — 27 citations

First Author

3. *JWST Observations of Starbursts: Dust Processing in the M82 Superwind*
   Cronin, Serena A., Bolatto, Alberto D., Richie, Helena M., Donnelly, Grant P., et al., 2026, ApJ, 1002, 217

2. *Physical Conditions of the Ionized Superwind in NGC 253 with VLT/MUSE*
   Cronin, Serena A., Bolatto, Alberto D., Congiu, Enrico, Donaghue, Keaton, et al., 2025, ApJ, 987, 92

1. *Local Environments of Low-redshift Supernovae*
   Cronin, Serena A., Utomo, Dyas, Leroy, Adam K., Behrens, Erica A., et al., 2021, ApJ, 923, 86

Significant Contributions

3. *The EDGE-CALIFA Survey: Star Formation Efficiency and Galaxy Quenching across 62 Main-sequence, Green Valley, and Red Galaxies*
   Teng, Yu-Hsuan, Bolatto, Alberto D., Teuben, Peter J., Rosolowsky, Erik, et al. (**incl. S. A. Cronin**), 2026, ApJ, 1007, 59

2. *JWST Observations of Starbursts: Polycyclic Aromatic Hydrocarbons Closely Trace the Cool Phase of M82's Galactic Wind*
   Lopez, Sebastian, Ring, Colton, Leroy, Adam K., **Cronin, Serena A.**, et al., 2026, ApJL, 999, L7

1. *JWST Observations of Starbursts: Polycyclic Aromatic Hydrocarbon Emission at the Base of the M82 Galactic Wind*
   Bolatto, Alberto D., Levy, Rebecca C., Tarantino, Elizabeth, Boyer, Martha L., et al. (**incl. S. A. Cronin**), 2024, ApJ, 967, 63

Contributing Author

12. *JWST Observations of Starbursts: The Motion of Dust-PAH Kinematics of M82 and NGC 253 with JWST/MIRI Spectroscopy*
    Donnan, Fergus R., Sandstrom, Karin, Bolatto, Alberto D., Draine, Bruce T., et al. (**incl. S. A. Cronin**), 2026, AJ, 172, 260

11. *JWST Observations of Starbursts: A Young Bubble in NGC 253's Central Starburst*
    Sheriff, Kaitlyn E., Mills, Elisabeth A. C., Mai, Xinyu, Siwakoti, Utsav, et al. (**incl. S. A. Cronin**), 2026, AJ, 172, 261

10. *JWST Observations of Starbursts: Molecular Hydrogen Excitation and Disequilibrium in M82*
    Duval, Sara E., Smith, J. D. T., Bolatto, Alberto D., Draine, B. T., et al. (**incl. S. A. Cronin**), 2026, ApJ, 1005, 56

9. *Building a culture of inclusion and allyship for queer astronomers*
   Brumback, McKinley, Calahan, Jenny, Cox, Isa, **Cronin, Serena**, et al., 2026, NatAs, 10, 764

8. *JWST Observations of Starbursts: Relations between PAH features and CO clouds in the starburst galaxy M 82 (Corrigendum)*
   Villanueva, V., Bolatto, A. D., Herrera-Camus, R., Leroy, A., et al. (**incl. S. A. Cronin**), 2025, A&A, 702, C2

7. *The Karl G. Jansky Very Large Array Local Group L-Band Survey (LGLBS)*
   Koch, Eric W., Leroy, Adam K., Rosolowsky, Erik W., Chomiuk, Laura, et al. (**incl. S. A. Cronin**), 2025, ApJS, 279, 35

6. *The MUSE view of the Sculptor galaxy: Survey overview and the luminosity function of planetary nebulae*
   Congiu, E., Scheuermann, F., Kreckel, K., Leroy, A., et al. (**incl. S. A. Cronin**), 2025, A&A, 700, A125

5. *The dust polarisation and magnetic field structure in the centre of NGC253 with ALMA*
   Belfiori, Davide, Paladino, Rosita, Hughes, Annie, Bernard, Jean-Philippe, et al. (**incl. S. A. Cronin**), 2025, A&A, 697, A24

4. *JWST observations of starbursts: cold clouds and plumes launching in the M 82 outflow*
   Fisher, Deanne B., Bolatto, Alberto D., Chisholm, John, Fielding, Drummond, et al. (**incl. S. A. Cronin**), 2025, MNRAS, 538, 3068

3. *JWST Observations of Starbursts: Relations between PAH features and CO clouds in the starburst galaxy M 82*
   Villanueva, V., Bolatto, A. D., Herrera-Camus, R., Leroy, A., et al. (**incl. S. A. Cronin**), 2025, A&A, 695, A202

2. *JWST Observations of Starbursts: Massive Star Clusters in the Central Starburst of M82*
   Levy, Rebecca C., Bolatto, Alberto D., Mayya, Divakara, Cuevas-Otahola, Bolivia, et al. (**incl. S. A. Cronin**), 2024, ApJL, 973, L55

1. *The EDGE-CALIFA Survey: Molecular Gas and Star Formation Activity across the Green Valley*
   Villanueva, Vicente, Bolatto, Alberto D., Vogel, Stuart N., Wong, Tony, et al. (**incl. S. A. Cronin**), 2024, ApJ, 962, 88